DO NOW — RECALL

# Assembly language and addressing modes

Name: _____ Class: _____ Date: _____

**1. Which primary memory is normally volatile?**

**2. What is the MAR used for?**

**3. What does an array index indicate?**

**4. For int[] a={5,8,3,7}, give a[2], then execute a[1]=a[1]+a[2]. State the new array and total.**

DO NOW — CHECK / OBJECTIVES

# Correct and improve

Check page 1 in GREEN PEN. Write corrections in full.

### 1. Model answer

RAM.

### 2. Model answer

Holding the address of memory being accessed.

### 3. Model answer

The position of an element within an array.

### 4. Model answer

a[2]=3; a[1]=11; array [5,11,3,7], sum 26.

### Vocabulary — know these terms

immediate — Operand value encoded as a literal.

direct — Memory address of data specified in instruction.

indirect — Specified memory holds a further effective address.

indexed — Base address modified by index or displacement.

effective address — Final memory location used for operand access.





# WE DO — SOURCE CHECK / IMPROVE

Only use this section AFTER attempting the We Do, You Do and Exit Ticket. Compare with the original classroom lesson's CHECK explanation. Correct errors in GREEN PEN and explain why.

### Guided 1 — original CHECK

Explain

M[41] points to address 42, so indirect reads 5. Indexed base 40+2 reads M[42]=5.

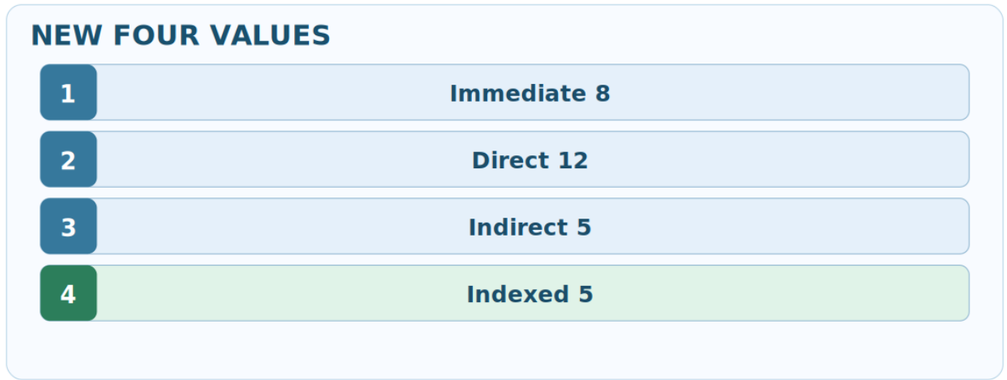
IMPROVE: Explain one correction to your own reasoning.

## We Do Check · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

**M[40]=12, M[41]=42, M[42]=5, X=2. Evaluate #8, [40], [[41]], [40+X].**

Immediate returns 8; direct reads M[40]=12.




NEW FOUR VALUES: Immediate 8; Direct 12; Indirect 5; Indexed 5





NEW LEARNING

## Read, annotate and explain

### Literal versus location

Immediate addressing embeds a literal value; direct addressing points to the memory address holding a value.

### Follow a pointer for indirect

M[21] contains 20, so an indirect read through address 21 accesses M[20]=7.

### Calculate indexed address

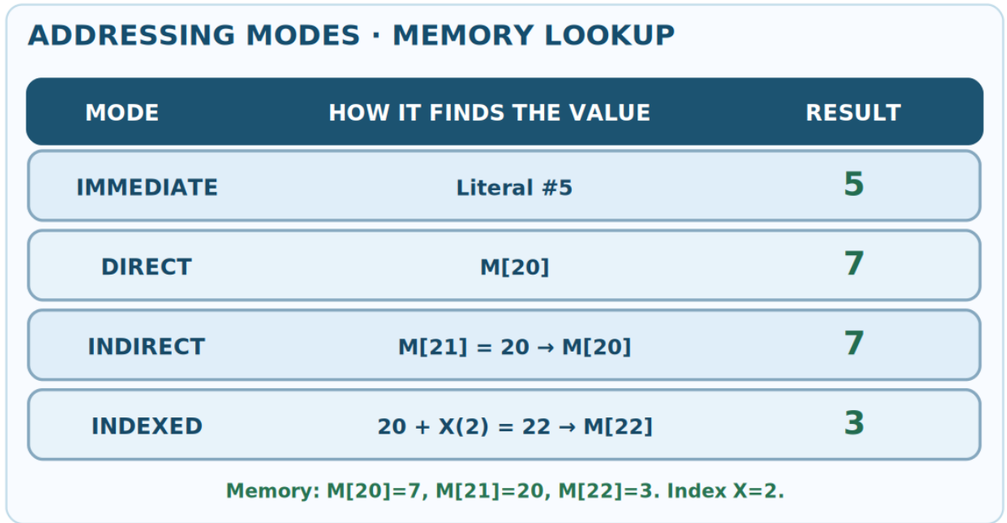
Base 20 plus index X=2 gives effective address 22 and value 3. These operand formats are generic examples rather than LMC syntax.

**Explain one step in your own words, using the correct technical terms.**

## New Learning · original lesson diagrams

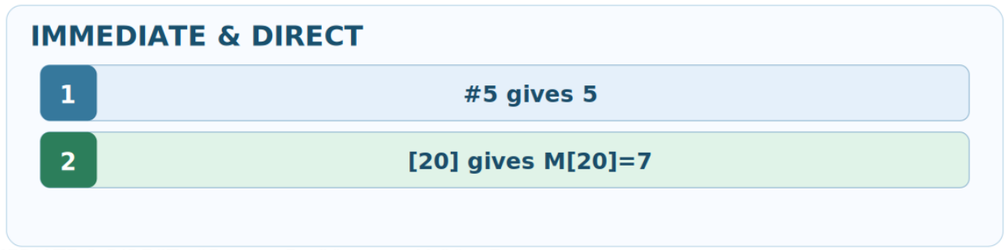
Diagrams appear after the related explanation. Annotate the original source visual.

**New Learning**




Immediate literal 5, direct M20 gives 7, indirect M21 points to M20 giving 7, indexed base 20 plus 2 gives value 3

**New Learning**




IMMEDIATE & DIRECT: #5 gives 5; [20] gives M[20]=7





EXIT TICKET / REFLECTION

## Check your understanding

Complete the actual lesson Exit Ticket before checking the answer.

**Indirect addressing returns the pointer value itself rather than following the pointer.**

**How confident are you? Circle one: Confident / Getting there / Need help**

Explain your choice, identifying one concept you can now use independently.

### GREEN PEN — Correct and explain

Choose one mistake from your independent work. Write the correct method and explain precisely why your original answer was wrong.

Next step: revisit the model or a relevant revision-guide section if needed.

YOU DO — INDEPENDENT PRACTICE

# Questions · Part 2

Answer independently. Show code, calculations, a trace or justified reasoning as appropriate.

### Question 3

An array of 16-bit values begins at byte address 100. Find byte address of array[3] and explain why base+3 would be wrong for byte-addressed memory.

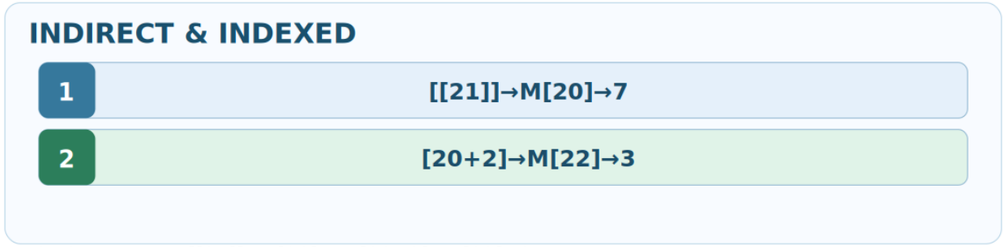




# New Learning · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

**New Learning**




INDIRECT & INDEXED: [[21]]→M[20]→7; [20+2]→M[22]→3

I DO — WORKED EXAMPLE

# Follow the worked method

**M[30]=4, M[31]=30, M[32]=9, X=2. Evaluate #6, [30], [[31]], [30+X].**

### Predict

Immediate is literal 6; direct gets 4 from memory address 30.

### Explain

Indirect follows pointer 30 stored at 31 to value 4; indexed adds offset 2 to base 30 to read 9.





YOU DO — INDEPENDENT PRACTICE

# Questions · Part 1

Answer independently. Show code, calculations, a trace or justified reasoning as appropriate.

### Question 1

Define immediate, direct, indirect and indexed addressing.

### Question 2

Given M[10]=3, M[11]=10, M[12]=9 and X=2, find #4, [10], [[11]] and [10+X], explaining both address calculations.

# WE DO — TRY · DISCUSS · IMPROVE

TRY: Use the original guided questions and diagrams in the preceding We Do pages. DISCUSS: Explain and challenge methods before independent practice. IMPROVE: Note one change now; compare with the source model only AFTER the Exit Ticket.

**Guided 1: Predict Immediate returns 8; direct reads M[40]=12.**

DISCUSS — Compare your methods. Justify one important decision, identify one difference, and agree a specific improvement.

IMPROVE — Write one precise correction or stronger explanation in your own answer.



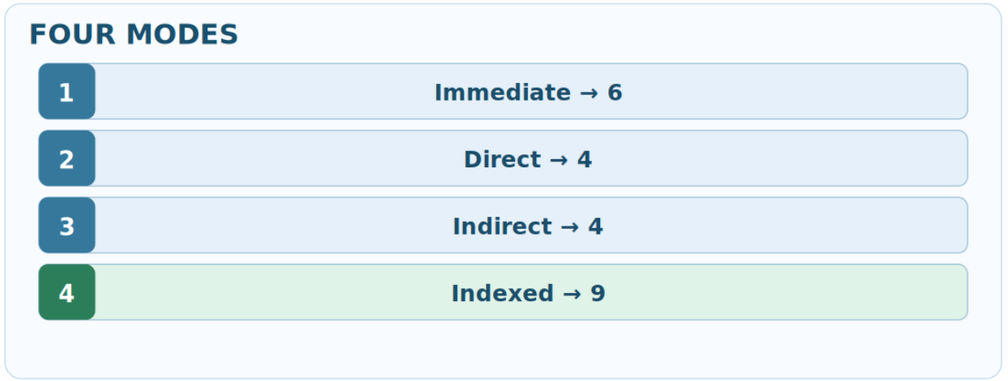


# I Do · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

**M[30]=4, M[31]=30, M[32]=9, X=2. Evaluate #6, [30], [[31]], [30+X].**

Immediate is literal 6; direct gets 4 from memory address 30.




FOUR MODES: Immediate → 6; Direct → 4; Indirect → 4; Indexed → 9

WE DO — GUIDED PRACTICE

# Try → Discuss → Improve



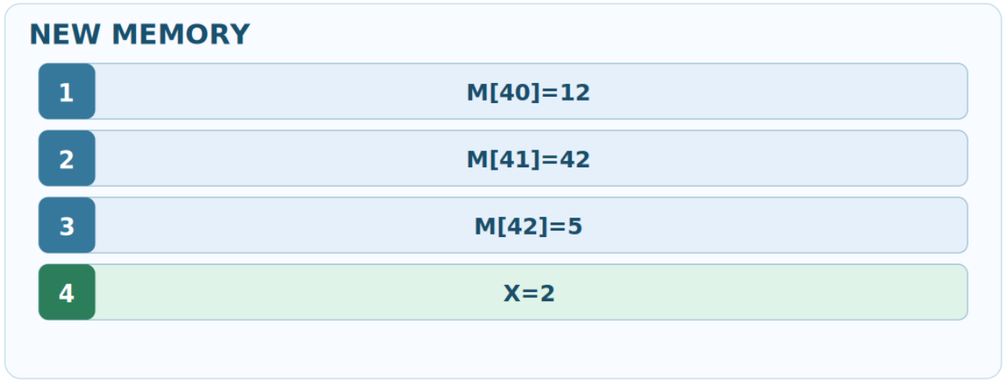


## We Do · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

**M[40]=12, M[41]=42, M[42]=5, X=2. Evaluate #8, [40], [[41]], [40+X].**

Immediate returns 8; direct reads M[40]=12.




NEW MEMORY: M[40]=12; M[41]=42; M[42]=5; X=2